



A Survey of Deep Learning Approaches for the Monitoring and Classification of Seagrass

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Received: 21 December 2024 / Revised: 11 March 2025 / Accepted: 1 April 2025 / Published online: 15 May 2025
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Abstract

Seagrass meadows are essential to the health of coastal ecosystems. They support carbon storage, provide habitats for marine species, and help stabilize coastlines. Monitoring underwater seagrass is vital for understanding the conditions of the ecosystem. Researchers have been interested in identifying and classifying underwater seagrasses. However, traditional monitoring methods can be labor-intensive and costly, especially in complex underwater environments. Deep learning approaches have made significant progress in digital image processing, particularly in object recognition and classification, and are among the most popular computer vision tools. The collection of digital images for monitoring underwater habitats, such as seagrass meadows, has increased significantly as recent progress in imaging technology has made it easier to collect high-resolution data. The increase in imagery data has in turn created a demand for automated detection and classification using deep neural network-based techniques. This study reviews the current deep-learning techniques used for monitoring and classification of the seagrass. It discusses the key methodologies, datasets, and progress in this area. This study not only examines the well-known challenges such as limited availability of data but provides a novel, structured taxonomy of deep learning techniques tailored for the monitoring of seagrass, highlighting their unique advantages and limitations within diverse marine environments. By synthesizing findings across various data sources and model architectures, we offer critical insights into the selection of context-aware algorithms and identify key research gaps, an essential step for advancing the reliability and applicability of AI-driven seagrass conservation efforts.

Keywords Seagrass · Deep neural networks · Machine learning · Deep learning · Classification · Monitoring · Underwater

1 Introduction

The ocean covers approximately 71% of the earth's surface (Hawthorne 2019), and seagrass meadows are vital for the environmental resilience of coastal regions. These meadows support marine food webs, provide shelter for numerous fish species, and contribute to nutrient cycling (Kelaher et al. 2013). Seagrasses are key contributors to coastal health; they produce oxygen and are a significant carbon sink (Fig. 1).

However, these ecosystems face growing threats from human activities, pollution, and climate change, which lead to the loss of habitats and reduced availability of light for photosynthesis (Beca-Carretero et al. 2024). A decline in seagrass habitats directly affects biodiversity, fisheries, and shoreline stability. Therefore, the monitoring of seagrass is essential for effective marine ecological management (Borum et al. 2004). It helps track ecosystems and health, supports conservation efforts, and enables the early detection of environmental changes, ultimately contributing to the preservation of these valuable underwater habitats.

Advancements in underwater digital imaging (Moghim and Mohanna 2021) with technologies such as autonomous underwater vehicles (AUVs), unmanned underwater vehicles (UUVs), digital cameras, video imagery, and satellite remote sensing techniques, have significantly enhanced the ability to capture underwater images. These technologies provide high-resolution data, enabling large-scale, detailed observations of seagrass meadows. However, conventional

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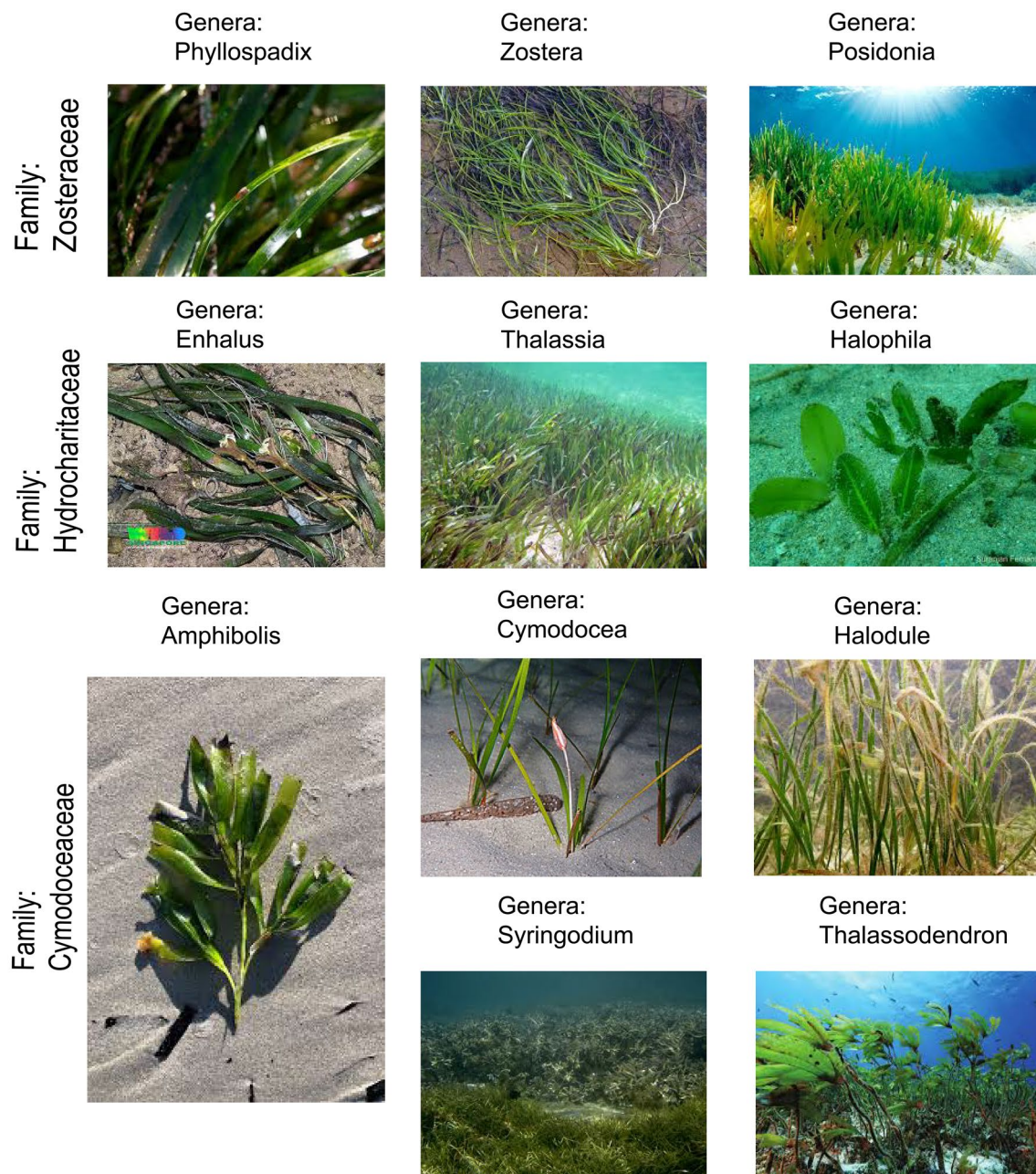


Fig. 1 Seagrasses meadows of different species. Adapted from Noman et al. (2021a)

methods of monitoring are labor-intensive, time-consuming, and often require expert analysis (Imhoff et al. 2011). The growing volume of underwater imagery has necessitated more efficient processing and interpretation. Artificial intelligence (AI), particularly deep learning-based algorithms (He et al. 2022; Halder et al. 2024), offers robust solutions to these challenges. These methods can automate the detection, classification, and mapping of seagrass, thereby reducing processing time significantly and increasing accuracy. Furthermore, AI-based approaches have enhanced our understanding of seagrass dynamics over time (Sultana

et al. 2024). While increasing the amount of data generally enhances model performance by providing more diverse training samples, the improvement is not always linear or guaranteed. Some studies show that additional data does not significantly improve accuracy beyond a certain threshold and may in fact introduce noise, which leads to diminishing returns. Factors such as data redundancy, label noise, and computational complexity can limit the effectiveness of large-scale datasets. For example, studies on classification of seagrass have shown that while initial performance gains are notable, excessive data without quality enhancements

can lead to model saturation. This suggests that rather than simply increasing data size, improving data quality, diversity, and augmentation techniques is essential to enhance the accuracy of classification.

Recently, machine learning (ML) methodologies have been utilized for classifying and monitoring seagrass ecosystems (Bakirman and Gumusay 2020). These approaches have demonstrated the potential to improve the precision and efficacy of seagrass mapping, which is vital for conservation initiatives. A hybrid machine learning approach, Random Forest, integrated with the Swarm Whale Optimization Algorithm (RF-SWOA) (He et al. 2023) has been designed to enhance the predictive accuracy and stability of habitat suitability models. This method combines the resilience of Random Forests with the optimization potential of the Swarm Whale Algorithm to improve the modeling of complex seagrass ecosystems. A comparative evaluation of machine learning algorithms is presented in (Thomasberger et al. 2023). Several machine learning algorithms have been explored for the classification of seagrass, including Support Vector Machines (SVM), Random Trees (RT), *k*-Nearest Neighbor (kNN), Bayesian techniques, and Decision Trees (DT). These algorithms categorize submerged seagrass species in images obtained from uncrewed aerial systems (UAS), indicating that machine learning can be applied to assess UAS data for aquatic vegetation efficiently. These findings underscore the increasing significance of machine learning approaches in seagrass monitoring and the potential to enhance the precision of classification and computational efficiency across various data sources, including aerial and underwater images. These innovations have enhanced the mapping and monitoring of habitats, bolstering seagrass conservation and ecosystem management. Although machine learning techniques are being applied extensively for the classification of seagrass, they face multiple limitations that affect their efficacy and scalability (Lisani et al. 2022). These traditional approaches rely on handcrafted features and statistical learning methods, which limit their ability to generalize across varying underwater conditions. One of the key limitations of machine learning in seagrass classification is its dependency on manually engineered features, which may not capture complex spatial patterns effectively. The main problem is the reliance on extensive high-quality labeled datasets, which are sometimes challenging to acquire in marine environments. The deficiency of such labeled data constrains the capacity of the model to form generalizations effectively, resulting in problems such as overfitting and poor performance on novel, unseen data. The necessity for human annotation of underwater imaging is laborious and time-consuming (Malik 2020), hence limiting the accessibility of datasets. Moreover, environmental factors, such as water turbidity, lighting, depth, and seasonal fluctuations, may

impact image quality significantly and, subsequently, the precision of models developed from static datasets (Pühr et al. 2014).

The complex nature of seagrass ecosystems, characterized by diverse species, thick vegetation, and overlapping meadows, poses additional challenges for precise classification (Gambin et al. 2021). Morphological differences among seagrass species and the presence of other coastal flora and animals complicate the process of identifying and classifying species based simply on visual indicators. Moreover, spatial and spectral variability in aquatic environments might result in misclassification because models may not be well-equipped to distinguish between seagrass and visually similar substrates, such as algae, coral, or sand. These challenges underscore the necessity for durable and flexible models for managing various marine ecosystems. Advancing deep learning methodologies and utilizing techniques such as transfer learning, data augmentation, and domain adaptation are essential for addressing these limitations and improving the reliability and effectiveness of seagrass monitoring.

There is an increasing demand for deep learning models for the monitoring of seagrass because of their exceptional capacity to handle large amounts of data and reveal intricate patterns in underwater images (Khan et al. 2020). Models such as Convolutional Neural Networks (CNNs) and other deep architectures have shown significant capability in extracting relevant features from raw data, thereby improving the accuracy of classification (Zhong et al. 2016). The primary advantage of deep learning is its capacity to acquire hierarchical and abstract representations, reducing the need for manual feature engineering. This results in more accurate and durable mapping of seagrass meadows, regardless of changing environmental variables, including shifting light, water turbidity, and seasonal alterations (Hossain and Hashim 2019). Moreover, deep learning models efficiently absorb multi-modal data from diverse sources, including satellite images, aerial images, and underwater films, thereby enhancing the efficacy of seagrass classification. Ongoing enhancements in processing capacity render these models increasingly accessible, facilitating real-time research and monitoring of large coastal regions. Consequently, deep learning presents a possible avenue for addressing issues in the monitoring of seagrass, enabling more efficient conservation and management measures.

Our survey of recent deep-learning algorithms for the classification of seagrass focused on techniques that use various data sources, including satellite imagery, video recordings, sensor-based data, and underwater multi-species datasets. We highlight the distinct challenges and improvements in each technique. The survey included journal and conference papers and book chapters found through web searches. We utilized the Web of Science,

Science Direct, Google Scholar, IEEE Xplore digital library, and Springer. We examined the performance of deep learning models for each data type. Previous studies Ozaeta et al. (2024), Noman et al. (2024), Menghani (2023) and Moniruzzaman et al. (2019) have explored seagrass classification, which is increasingly affected by rapid environmental changes due to climate change and human activities. This paper addresses the significant challenges posed by ecological changes driven by climate change and human activities, which makes the classification of seagrass increasingly complex and crucial for conservation efforts.

Given the significant improvements in deep learning applications for the monitoring of seagrass, this study aims to thoroughly analyze existing methodologies, datasets, and challenges related to seagrass classification. The primary goal is to assess the effectiveness of deep learning models in automating seagrass detection and the classification of species while identifying significant limitations that affect model performance. Furthermore, this study investigates how diverse data sources, such as satellite imaging, underwater imagery, and sensor-based data, help to improve the accuracy of classification. While previous surveys have offered overviews of seagrass monitoring techniques, this paper distinguishes itself by providing a focused, in-depth review of deep learning systems that have been specially designed to handle the unique problems of aquatic settings. Our survey is more than just a compilation of methods; it includes a structured taxonomy that categorizes deep learning models based on their architecture, data requirements, and performance characteristics. It provides a practical guide for researchers and practitioners in selecting the best techniques for their specific monitoring needs. Furthermore, we combine findings from other data sources—satellite imaging, underwater imaging, and sensor-based data—to draw new insights into how different data streams might be linked synergistically to increase the accuracy and robustness of seagrass classification. This comprehensive approach and critical assessment of current models' limitations distinguish this work and place it as an essential resource for advancing AI-driven seagrass conservation efforts.

The remainder part of the paper is organized as follows: Sect. 2 reviews related surveys. Section 3 compares traditional machine learning and deep learning approaches used for seagrass classification. Section 4 outlines the criteria for paper selection. Section 5 provides a comprehensive overview and taxonomy of deep-learning techniques. Section 6 details data acquisition, and Sect. 7 discusses data preparation. Section 8 describes detection approaches, and Sect. 9 focuses on learning methods. Section 10 presents the deep learning architectures. Performance evaluation metrics are discussed in Sect. 11. Section 12 explores the challenges faced by the various approaches. Section 13 provides a

discussion, and Sects. 14 and 15 present the conclusions and the direction for future research.

2 Related Survey

Machine learning and deep learning techniques have been used extensively to detect, monitor, and classify seagrasses. Moniruzzaman et al. (2019) evaluated the current methods for the recognition and mapping of seagrass to identify gaps and outline future research possibilities for enhanced monitoring of ocean health. Initial monitoring methods depended on in situ observations by experts; however, innovations in remote sensing, satellite imaging, digital imagery, and videography have facilitated more rapid and precise approaches. An analytical study identified deficiencies in current research methods, including the necessity for economical and efficient imaging methodologies resilient to environmental variables, such as depth, turbidity, and weather, with completely automated real-time seagrass recognition and precise density estimation. This research indicates that further investigations should prioritize the creation of cost-effective data-gathering techniques. Noman et al. (2021a; b), introduced a semi-supervised two-stage framework for seagrass classification. The authors employed a limited quantity of labeled data and an extensive collection of unlabeled data. An EfficientNet Convolutional Neural Network (CNN) was initially trained using annotated seagrass image segments. The trained classifier subsequently produces pseudo labels for the unlabeled patches. Next the classifier is retrained using both the original labeled and pseudo-labeled data. The approach attained a multi-species detection accuracy of 98.0% on a newly developed seagrass dataset. The publicly available 'DeepSeagrass' dataset produced accuracies of 91.3% and 93.0% for four and five classes, respectively, outperforming previous best values of 88.2% and 92.4%. Ozaeta et al. (2024) developed an innovative deep-learning approach to enhance the classification of seagrass species by integrating image-based data with an extensive dataset encompassing temporal fluctuations, varied geographic locations, water depths, and turbidity levels. An early stopping differentiable architectural search was employed to train the neural network, and a classification accuracy of 93.72% was obtained by this method. This study focused on five seagrass species. This study highlights the efficacy of deep learning algorithms in addressing issues associated with the complex characteristics of seagrass species. Yamakita et al. (2019) investigated the application of image classification and deep learning techniques to monitor seagrass meadows in the Hat Chao Mai National Park, Thailand. They used historical aerial images from the 1970s in their investigation. Despite the low accuracy in classifying different densities of

plant cover, both supervised and deep learning techniques detected the presence of seagrass effectively with an accuracy of 80%. Even when utilized on grayscale images, deep learning provides an accuracy equivalent to or greater than conventional classification methods. This study demonstrates the efficacy of these techniques for monitoring the temporal dynamics of seagrass regions and observing changes within specific sub-regions, particularly next to river mouths. Tahara et al. (2022) examined several image classification methodologies for mapping multi-specific seagrass beds in the Saroma-ko Lagoon, Hokkaido, Japan. Owing to the constraints of low-resolution satellite and aerial imagery in identifying species composition, this study utilized high-resolution UAV imagery to examine detailed alterations in seagrass. They evaluated three techniques: pixel-based classification, object-based classification, and deep neural networks (DNN). The DNN methodology attained the best accuracy (82%) in classifying seagrass beds and species composition. Seagrass meadows are an essential component of coastal ecosystems; however, they are constantly being damaged by human activity. Regular and effective monitoring of seagrass meadows is widely acknowledged as essential for their protection since several studies have pointed to excessive human impact on marine ecosystems. Scarpetta et al. (2022) investigated the use of a deep-learning-based image segmentation algorithm for monitoring coastal seagrass meadows using satellite imagery. The model can classify pixel-wise images by identifying the land, ocean, and seagrass meadows. A dataset of high-resolution satellite images of areas around the Apulian coastline was generated and utilized as the training dataset for the deep learning network. The classification ability of the model was evaluated using a collection of test images, which provided highly encouraging findings.

3 Traditional ML Versus DL-Based Seagrass Classification Techniques

A conventional machine learning methodology for the classification of seagrass includes the following important stages: image acquisition, pre-processing including image enhancement, extraction or selection of features, implementation of a machine learning classifier, and performance evaluation (Traganos and Reinartz 2018; Ha et al. 2020; He et al. 2023). Various image processing techniques have been employed for detection of seagrass, including the extraction of form, color, and texture attributes. Although shape features help identify seagrass structures, color features are frequently affected by water quality and light conditions (Duffy et al. 2018; Xu et al. 2023). Conversely, textual features are essential for differentiating seagrass from other benthic ecosystems (Mohamed et al.

2020a, b). A significant problem in the classification of seagrass is the apparent resemblance between many seagrass species and other underwater substrates, which might compromise the accuracy of the models.

Machine learning algorithms often depend on labeled training data to identify unique features. Traditional methodologies frequently require the integration of many data modalities, including shape, texture, and spectral information, to enhance their accuracy. Some researchers have integrated color and texture features to enhance the capacity for differentiation of seagrass species, whereas others have employed spectral data for improved discrimination (Hossain et al. 2015). Advanced methodologies have also derived features utilizing a gray-level co-occurrence matrix (GLCM), spectral descriptors, and morphological characteristics to improve the efficacy of classification. Algorithms, such as Support Vector Machines (SVM), Random Forests, and other feature selection techniques, have been used for their efficacy in the classification of seagrass.

Deep learning (DL) substantially benefits conventional machine learning by autonomously acquiring relevant characteristics from unprocessed data and eliminating the need for manual extraction of features (Ahmed et al. 2023). Deep learning methods, such as Convolutional Neural Networks (CNNs), have demonstrated significant potential for image classification applications, including seagrass mapping. They may be equipped to examine complex patterns in underwater images using hierarchical feature learning, in which higher level features are derived from lower-level qualities (Li et al. 2024; Chen et al. 2024). Recurrent Neural Networks (RNNs) are utilized for tasks requiring temporal data analysis, such as observing seasonal variations in seagrass coverage.

Various network architectures are popular for deep learning in the identification of seagrass. Convolutional Neural Networks (CNNs) are widely used because of their capacity to filter and analyze spatial patterns in images. On contrast, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are advantageous for temporal dynamics. Depending on the network design, components such as convolutional layers, pooling layers, activation functions, and memory cells are utilized for effective learning of features and classification (Namatēvs 2017).

Deep learning models typically outperform conventional methods owing to their robust skills for feature extraction and capacity to manage extensive datasets. Traditional machine-learning approaches frequently reach a performance plateau as data quantity expands. However, deep learning models enhance their performance consistently with additional data (Najafabadi et al. 2015). This has resulted in a growing preference for deep learning in the monitoring of

seagrass, which is essential for precise classification of the species, habitat mapping, and tracking of temporal change. Figure 2 presents a conceptual taxonomy, which categorizes different aspects such as data acquisition, image pre-processing, learning methods, and deep learning architectures. This hierarchical representation helps in understanding the broad structure of the workflow. However, not all stages are necessary for every methodology. This study outlines the approaches used in several studies for deep learning-based classification of seagrass, offering insights into data preparation, model selection, and evaluation procedures to enhance the monitoring and preservation of seagrass environments.

4 Criteria for Paper Selection

To assess the current status of research on seagrass monitoring using deep learning, we conducted a thorough literature review that included two steps: (i) identifying and choosing relevant studies and (ii) conducting an in-depth analysis of these studies. The critical research issue in directing this study was: What function can deep learning approaches serve in detecting, localizing, and classifying seagrass species? To gather relevant studies

for our research, we searched keyword-based databases, including Google Scholar, Web of Science, IEEE Xplore, Scopus, ScienceDirect, Multidisciplinary Digital Publishing Institute (MDPI), Springer, and university library resources. The research concentrated on journal articles and conference papers published from 2017 to 2024. After collecting documents from various databases, we removed duplicates and focused on research using deep learning approaches. Publications on this subject were limited before 2017; however, a significant rising trend has been observed in recent years. Therefore, our review mainly focuses on works published from 2017 onwards to represent the most recent developments in the field.

5 A Comprehensive Overview and Taxonomy of Deep Learning Techniques for Seagrass

Figure 3 provides a workflow-oriented perspective, which emphasizes a step-by-step process from data collection to model evaluation. It integrates essential operations such as image resizing, augmentation, and evaluation metrics, which are important for practical implementation. Critical

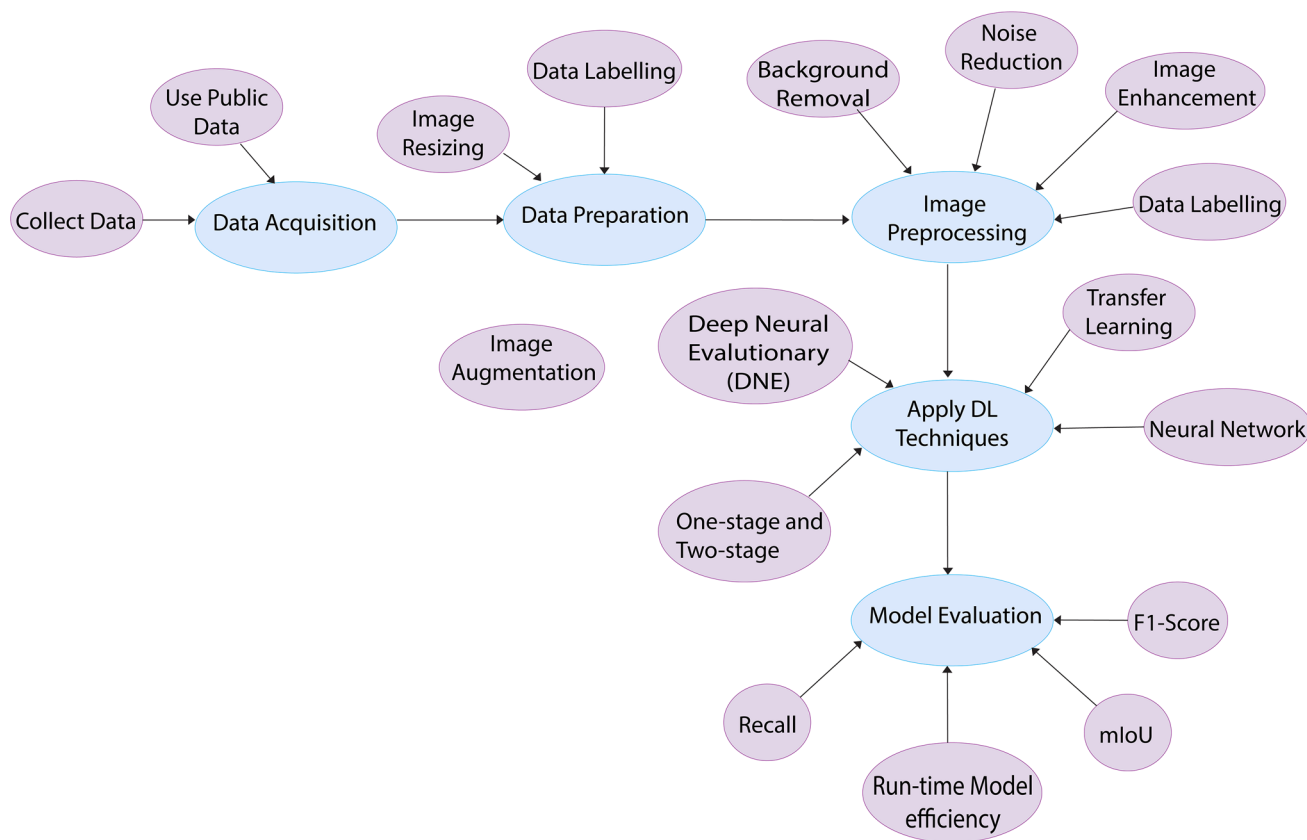


Fig. 2 An overview of the process for seagrass detection methods using deep learning approaches

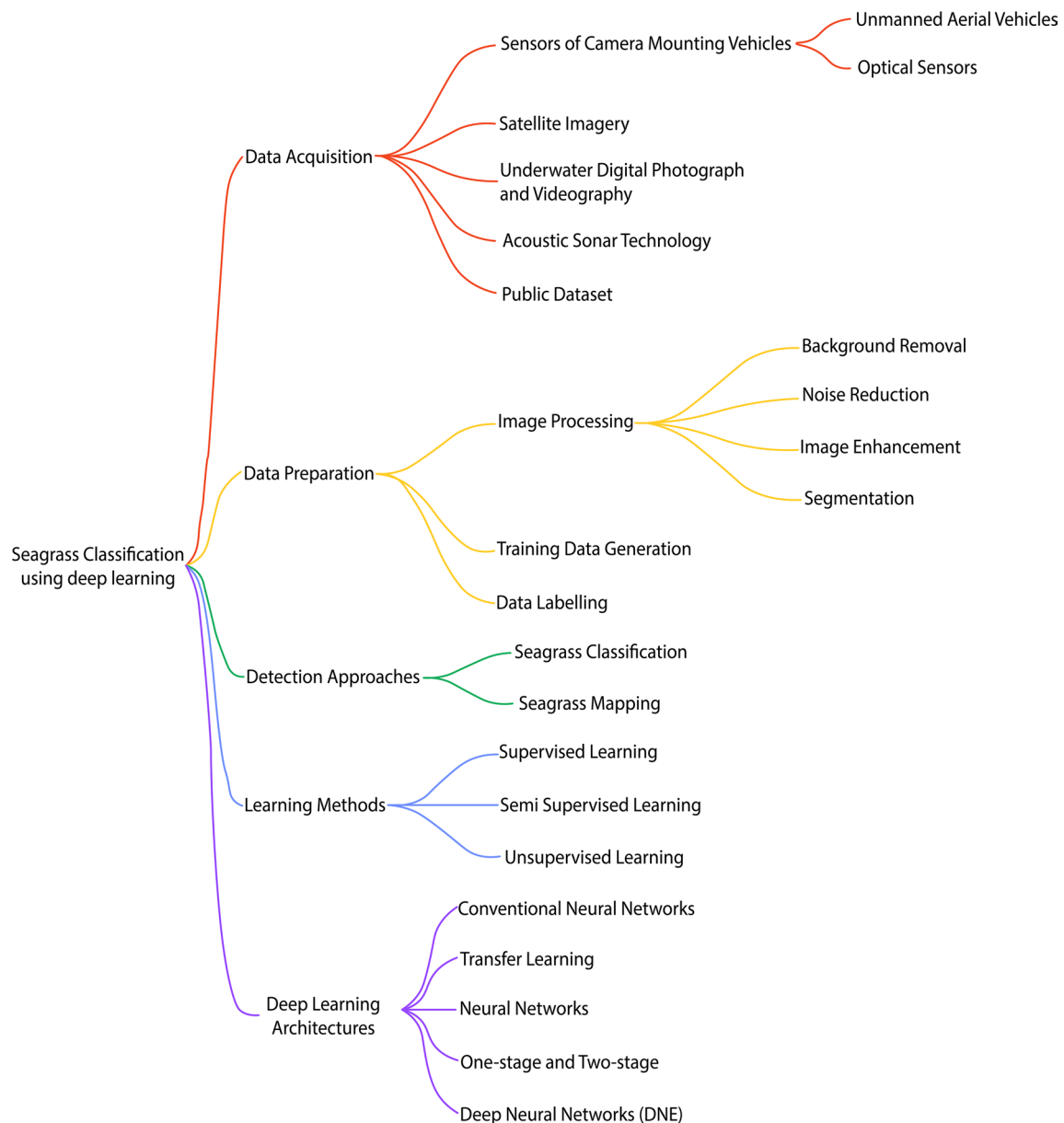


Fig. 3 An overall work-flow-oriented deep learning-based seagrass detection technique

elements, including data-gathering techniques, sensor types, and mounting platforms, are emphasized. A summary of the methodologies employed for the preparation of datasets, including image pre-processing, data production, and annotation, is presented in Table 1. The evaluated works produced seagrass distribution maps for certain regions or categorized seagrass species within a specific ecosystem. Researchers utilized supervised, unsupervised, and semi-supervised learning methodologies to create classification models contingent on the study objectives and data attributes. Diverse deep learning architectures have been employed based on these methodologies.

6 Data Acquisition

Techniques for seagrass detection and classification based on deep learning require an adequate quantity of labeled data. Various sensors mounted on diverse platforms have been used to acquire diverse data modalities. We examine the most common techniques used for seagrass data collection.

Table 1 An overview of deep learning approaches for seagrass detection

References	Dataset	Deep learning technique applied	Operation performed
Martin-Abadal et al. (2018)	Proprietary seagrass dataset	VGG16-FCN8	Segmentation and classification
Weidmann et al. (2019)	LookingForSeagrass	DeepLabV3+, FCN, dilNet, U-Net	Comparison of deep learning models
Wang et al. (2020)	LookingForSeagrass	Encoder-Decoder segmentation model	Segmentation with high mIoU
Noman et al. (2023)	Proprietary seagrass and DeepSeagrass datasets	YOLOv5-based one-stage, EfficientDetD7-based	Detection and classification of seagrass
Moniruzzaman et al. (2019)	Underwater <i>Halophila ovalis</i> images	Faster R-CNN (Inception V2 backbone)	Detection of <i>Halophila ovalis</i> leaves
Noman et al. (2021b)	Underwater <i>Halophila ovalis</i> images	Faster R-CNN (NASNet and EfficientDet-D7)	Improved faster R-CNN accuracy
Raine et al. (2020)	DeepSeagrass	VGG-16, Resnet-50	Patch-based classification using transfer learning
Noman et al. (2021a)	DeepSeagrass and ECU-MSS	EfficientNet-B5	Semi-supervised labeling and classification
Burguera (2020)	Posidonia oceanica dataset	Custom CNN-based models	Mapping seagrass covers and species
Jeon et al. (2021)	Custom drone imagery	SegNet, U-Net, PSPNet, DeepLab v3+	Monitoring habitats using drone imagery
Noman et al. (2024)	DeepSeagrass	Boosted Atomic Orbital Search (BAOS) with CNNs	Optimization of CNN architectures and hyperparameters
Paul et al. (2023)	DeepSeagrass	Lightweight DeepSeagrass (LWDS)	Classification of seagrass using deep learning technique
Loureiro et al. (2024)	Galicia coast seabed dataset	CNN-SVM model	Seabed characterization and habitat mapping
Ozaeta et al. (2024)	Seagrass species classification dataset	Differentiable Architecture Search (DARTS)	Classification of five seagrass species from Calatagan, Batangas, and Puerto Galera, Oriental Mindoro, Philippines
Lestari et al. (2021)	Seagrass <i>Enhalus acoroides</i>	Mask R-CNN	Detection of seagrass
Tahara et al. (2022)	Saroma-ko-Lagoon seagrass dataset	Deep neural network (DNN)	Mapping multi-specific seagrass
Scarpetta et al. (2022)	Apulian coastline satellite image dataset	U-Net model	Segmentation using deep learning approaches
Mohamed et al. (2020b)	Shiraho and Fukido seagrass dataset	CNN	Detecting and mapping seagrass species
Raine et al. (2024)	DeepSeagrass	SeaFeats and SeaCLIP	Segmentation of seagrass
Tallam et al. (2023)	High-resolution drone imagery	DeepLab v3	Image segmentation and classification of seagrass
Marre et al. (2020)	Underwater photogrammetry data taken from 21 study sites	Use of automatic classification and reconstruction uncertainty	Automatic classification of seagrass patches
Perez et al. (2020)	8-band satellite images	CNN and Deep Capsule Network (DCN)	Classification and regression tasks for seagrass distribution

6.1 Sensors and Camera Mounting Vehicle

6.1.1 Unmanned Aerial Vehicles (UAVs)

Unmanned Aerial Vehicles (UAVs) are often used for data collection in seagrass research. UAVs offer an effective method for assessing the density and distribution of seagrass across extensive coastal regions by acquiring high-resolution RGB or multispectral imagery (Elma et al. 2024; Karang et al. 2024); . These images examined seagrass meadows, helping in the identification of species taxonomy, evaluation of habitat health, and temporal coverage changes. Unmanned

Aerial Vehicles (UAVs) are highly effective in detecting seagrass beds and differentiating them from adjacent environments, including sand and other benthic flora. UAVs can cover large regions in a single flight by operating at different heights and creating detailed seagrass maps (Yang et al. 2020a, b; Price et al. 2022). Numerous studies have partitioned images into smaller patches to enhance thorough analysis and augment the accuracy of classification among seagrass species and other marine vegetation (Ventura et al. 2018). For detailed monitoring, UAVs can operate at reduced altitudes (approximately 2–2.5 m), obtaining sufficient imagery to identify and classify individual seagrass

patches (Veettil et al. 2020). Flight settings, including altitude, camera angle, and overlap sequences, were carefully regulated to provide excellent coverage and quality. For example, flying between 10 and 20 m at altitudes balances coverage and image resolution. Simultaneously, overlaps between successive images are thoroughly collected in the research area without errors. Furthermore, advanced UAVs equipped with RGB and multispectral cameras can acquire images with resolutions of about 3000×4000 pixels or greater, allowing a detailed examination of seagrass beds and their present conditions.

6.1.2 Optical Sensors

Monitoring using optical sensors is an effective method for mapping seagrass and evaluating its health. These sensors are often affixed to platforms such as boats, underwater remotely operated vehicles (ROVs), or unmanned aerial vehicles (UAVs) to acquire spectral data that provide comprehensive insights into seagrass distribution, species composition, and physiological conditions. Optical sensors function across various spectral bands, including the visible, near-infrared, and hyperspectral ranges, facilitating the identification of complex variations in seagrass reflectance linked to distinct species, chlorophyll concentration, and degrees of stress or nutrient availability. By analyzing spectral signatures from these sensors, researchers can effectively distinguish between seagrass and other benthic substrates such as algae, coral, and sand, facilitating precise classification of the habitats (Loureiro et al. 2024). Furthermore, optical sensor data can facilitate the monitoring of seagrass meadow health and density by evaluating characteristics such as biomass, canopy structure, and coverage. This method facilitates swift and non-invasive acquisition of data across extensive regions, rendering it especially beneficial for sustained monitoring and conservation efforts in diverse coastal ecosystems.

6.2 Satellite Imagery

Satellite imagery is essential for large-scale monitoring of seagrass, as it provides a cost-efficient method for monitoring and evaluating seagrass meadows over vast coastal regions (Pu et al. 2010). Satellites equipped with multispectral and hyperspectral sensors acquire images across many spectral bands, thereby providing important information on seagrass distribution, species diversity, and habitat changes. High-resolution satellite imagery facilitates the mapping of seagrass meadow distribution and the evaluation of temporal changes (Mederos-Barrera et al. 2022), enabling investigations on ecosystem health and the influence of environmental variables. Multiple satellites, including Landsat (Topouzelis et al. 2018), Sentinel-2 (Bayyana et al.

2020; Ha et al. 2020), and WorldView (Baumstark et al. 2016; Manuputty et al. 2016), offer multispectral images of varying resolutions, allowing comprehensive mapping of seagrass habitats and regional and global evaluations. Images can be processed and analyzed to differentiate seagrass from other coastal elements such as sand, coral reefs, and algae.

A primary benefit of satellite imagery is its capacity to provide temporal insights, allowing the monitoring of seagrass patterns over many seasons and years. This extended observation is essential for understanding the impacts climate change, pollution, coastal development, and natural catastrophes have on seagrass meadows (Phinn et al. 2018). However, challenges exist in utilizing satellite imaging for the monitoring of seagrass, such as water turbidity, cloud cover, and seagrass bed depths, which can influence the clarity and precision of images. Advancements in image processing methods, data fusion approaches, and deep learning classification have enhanced the accuracy of seagrass detection and categorization in satellite imagery (Barrell et al. 2015). Consequently, satellite-based monitoring of seagrass is becoming progressively dependable and efficient for protecting and managing essential coastal ecosystems.

6.3 Underwater Digital Photography and Videography

Underwater imaging and videography are effective instruments for monitoring of seagrass, delivering high-resolution data that capture the delicate specifics of seagrass ecosystems (Marre et al. 2020). Researchers can collect exact visual data on seagrass beds, including species composition, density, and health, using cameras mounted on underwater platforms such as remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs), divers, or fixed frames (Islam et al. 2017). Videos and images provide unique perspectives of the underwater ecosystem, enabling the accurate identification of seagrass species, evaluation of growth trends, and examination of biological interactions within the habitat. Advancements in imaging technology have enabled underwater cameras to deliver high-quality movies to analyze seagrass biomass, canopy structure, and geographic distribution (Papakonstantinou et al. 2020). Videography facilitates continuous acquisition of data by documenting the dynamic characteristics of seagrass meadows, which is crucial for observing the alterations induced by tidal fluctuations, currents, and interactions with marine life. Underwater video data may be acquired while navigating extensive areas of seagrass beds, providing thorough coverage that static images alone cannot provide (Roelfsema et al. 2009). These data are advantageous for real-time evaluations of habitat health and monitoring

the impacts of environmental disturbances such as storm occurrences, anchor damage, and human activities. Videos facilitate the extraction of geographical and temporal data, offering insights into the dynamics of seagrass meadows over time and under varying conditions (Norris et al. 1997; Haag et al. 2008). Processing underwater images and videography presents challenges owing to fluctuating light conditions, water turbidity, and diverse marine flora and wildlife (Rende et al. 2015). Advancements in image enhancement and analysis techniques, particularly deep learning algorithms, have markedly enhanced the capacity to extract relevant information from such data (Halder et al. 2024). Methods such as object identification, segmentation, and pattern recognition have enabled the automated categorization of seagrass species and precise mapping of their distribution. Mosaicking techniques, which involve stitching multiple images or video frames together, allow the production of extensive maps of seagrass meadows, offering a comprehensive perspective of the environment. Underwater imaging and videography are essential for comprehensive monitoring of seagrass and for facilitating conservation and management efforts (Hafizt et al. 2024).

6.4 Acoustic Sonar Technology

Acoustic sonar technology is an effective technique for identifying seagrass beds, particularly in turbid waters where optical imaging is difficult. Sonar systems, including side-scan and multi-beam sonar, produce sound waves that traverse water and reflect off the bottom, producing complex audio images of underwater landscapes (Gumusay et al. 2019). Acoustic reflections differ according to the substrate and vegetation type, facilitating the identification of seagrass meadows and distinguishing them from sand, rock, or other benthic elements. This method facilitates massive, quick data acquisition across vast coastal regions and is especially proficient in identifying the bed density, coverage, and structure of seagrass (Rende et al. 2020). Although sonar mapping may not achieve detailed species-level categorization attainable by optical methods, it is significantly beneficial for extensive evaluation and monitoring of habitats, particularly in difficult situations where visual techniques are less efficient.

6.5 Public Dataset

Seagrass datasets are important for the monitoring of seagrass meadows. They enable researchers to analyze species distribution, health, and the impacts of environmental changes through satellite imagery, field surveys, and remote sensing data. The DeepSeagrass dataset is a multispecies dataset that is publicly available. It has 42,848 training, 10,720 validation, and 13,378 test image

patches (66,946 image patches). DeepSeagrass categorizes seagrass patches into four types: 'Background' (less than 1% of total seagrass), 'Ferny' (*Halophila spinulosa*), 'Rounded' (*Halophila ovalis*), and 'Strappy' (*Cymodocea serrulata*, *Halodule uninervis*, *Syringodium isoetifolium* and *Zostera muelleri*). The 'Background' class is further separated into 'Water' and 'Substrate' classes. Remote underwater video cameras acquired Global Wetlands data from Moreton Bay. The authors (Raine et al. 2024) evaluated the 764 'novel-test' split images, which were acquired from a different location than the 'train' and 'test' splits. To generate 38,200 test patches, authors (Raine et al. 2024) made 50 (10 × 5) patches from each image and manually labeled them as 'fish,' 'seagrass,' and 'background.' The seagrass in this dataset matches the 'Strappy' morphotype in the DeepSeagrass dataset.

ECU-MSS-2 is a new multi-species seagrass dataset created using underwater digital photographs. The 'ECU-MSS-2' dataset covers four habitats: 'Amphibolis,' 'Halophila,' 'Posidonia,' and 'Background.' The authors (Noman et al. 2021a; b) collected images from many sources. The Center for Marine Ecosystems Research (CMER) at Edith Cowan University in Western Australia gathered 'Halophila.' Images were taken at several sites in Western Australia, including Fremantle, Mandurah, Rottnest Island, Dampier Reserve, Pelican Point, Rocky Bay Swan River, Milyu, and Lucky Bay. Images were captured using a Fujifilm X-T30 mirror-less digital camera and a wXF 18–55 mm F 2.8–4 R LM OIS in natural light. The Department of Biodiversity, Conservation, and Attractions (DBCAs), Australia, gathered images contained in 'Amphibolis,' 'Background,' and 'Posidonia' from various marine parks in Western Australia, including Shark Bay, Shoalwater Island, Jurien Bay, Ngari Capes, Cockburn Sound, and Marmion. The images were taken from a downward-facing camera one meter above the seagrass canopy, with a field of view of 75 × 60 cm along the transects used to measure density/height. The dataset comprises 5201 images divided into four categories: 1304 'Amphibolis,' 1237 'Background,' 1315 'Halophila,' and 1345 'Posidonia.' It was separated into training and testing sets. The training set consists of 4161 images, divided into four categories: 'Amphibolis,' 'Background,' 'Halophila,' and 'Posidonia.' The test set included 1040 images, uniformly dispersed over four classes of 260 images each.

7 Data Preparation

After collecting data from several sources for the classification of seagrass, it is important to preprocess the dataset to train, test, and validate the deep learning models (Meister and Qu 2024). Raw data often requires significant

preparation before it can be directly used in a model (Mumuni and Mumuni 2024). This procedure includes the use of various image processing techniques, data labeling, and image augmentation to expand the dataset and introduce variability (Naveed et al. 2024)—through the generation of synthetic data to improve model training. Typical image preparation procedures for seagrass classification include the removal of background to separate seagrass characteristics, increasing image to fit model requirements, segmentation of particular components (such as green for seagrass flora), and removal of artifacts such as motion blur (Archana and Jeevaraj 2024). Additional methods include de-noising to enhance picture clarity, augmenting image quality, extracting vegetation indices for improved representation of features, and modifying the color model to improve classification performance. Ribeiro and Gonzaga (2006) analyzed video material by transforming it into sequences of RGB images and subsequently converting them to grayscale to emphasize key elements. Lønne (2014) utilized the auto-capture mode on their camera to acquire images in TIFF format, which were later transformed into the RGB color model to maintain consistency of data. In several instances, changing the image format proved essential for enhancing model training, mainly when using public datasets. The preprocessing stages guarantee that the seagrass data are organized optimally for training the deep learning model, enhancing the efficacy of the classification and detection tasks.

7.1 Image Pre-processing

Image processing is important in preparing seagrass data for deep learning models. It involves techniques like background removal, noise reduction, and image enhancement to improve visibility and focus on relevant features. Segmentation is crucial for identifying the components of seagrass and ensuring compatibility with model requirements. These processes help enhance data quality and enable more accurate classification and detection.

7.1.1 Background Removal

Background removal isolates the subject, such as seagrass, by removing extraneous elements, such as water or sand, from the image. This is accomplished by thresholding or machine learning methodologies such as the Normalized Difference Vegetation Index (NDVI), which focuses on vegetation features. Recent studies have presented novel methodologies for background reduction techniques in deep learning. Kang and An (2021) introduced an automated method for background removal that utilized deep learning-based depth estimation for digital image stitching. Their work focused on eliminating undesirable backgrounds from close-up scans of structure exteriors with Unmanned Aerial

Vehicle (UAV) footage, which is particularly relevant for monitoring seagrass in aquatic situations. Bici et al. (2022) used a Convolutional Neural Network (CNN) utilizing the U-NetMobilenetV2 architecture to automate background removal in photogrammetry applications. The CNN effectively distinguishes between the foreground (state pixels) and background, utilizing a dataset that includes over 700 RGB images. The Dice coefficient assesses performance, and the CNN-based method reduces the background removal time by more than 50% relative to manual techniques. The results were corroborated by 3D model reconstruction and juxtaposition with models acquired using a 3D scanner and conventional photogrammetry.

7.1.2 Noise Reduction

Noise reduction involves removing undesirable artifacts or distortions in the image, such as sensor noise or underwater particulates, to enhance precision in the classification of seagrass. Gaussian blur or CNN-based filters allow image smoothing while preserving the essential information. Cherian et al. (2021) developed a deep-learning technique for removing noise and haze from underwater imagery. A trigonometric filter with a complexity of $O(N)$ was used, where the input image was initially transformed into its frequency domain and subsequently treated with a high-pass filter to augment details. A convolutional neural network (CNN), consisting of three convolutional layers, was used to remove noise. The first layer processes a $176 \times 320 \times 3$ input image, with each subsequent layer using a convolution utilizing a 9×9 filter size, determined by the image's sigma (σ) using a bilateral trigonometric approach. Jin and Liang (2017) removed the noise from an image while maintaining its fundamental features. The conventional median filtering method calculates the median of the pixel values inside a sliding window and replaces the original value of the central pixel with the median value. This helps reduce the noise.

7.1.3 Image Enhancement

Image enhancement augments the visual quality of an image by modifying contrast, brightness, and sharpness, making it easier to distinguish objects such as seagrass from their background. This phase is crucial for underwater imagery, where lighting and visibility are frequently inadequate. Ancuti et al. (2012) proposed a fusion-based technique to enhance underwater images by integrating a contrast-enhanced image with a color-corrected variant using a multi-scale fusion approach. Subsequently, they enhanced this methodology by combining two images from a white-balanced underwater image with matching weighted maps, successfully restoring diminished features and edges. Fu et al. (2014) proposed a two-step methodology

for the improvement of underwater images, including a color correction technique utilizing piecewise linear transformation and a contrast enhancement algorithm. Fu et al. (2017) suggested an alternative solution using a Retinex-based approach for enhancement, including color correction, layer decomposition, and image enhancement. Zhang et al. (2017) enhanced the conventional Retinex technique by implementing bilateral and trilateral filters on image channels inside the CIELAB color space. Drews et al. (2013) modified the dark channel prior (DCP) for underwater environments, introducing the underwater dark channel prior (UDCP). Peng et al. (2018) introduced the generalized dark channel prior (GDCP), which improves images by including adaptive color correction in the image creation model. Li et al. (2018) generated underwater images from RGB-D data captured in the air and developed a two-stage network for image restoration utilizing the synthetic dataset. Li et al. (2019) also used a gated fusion network that generates three confidence maps to integrate three input images into one improved output. Ye et al. (2019) introduced an unsupervised adaptation network designed to estimate scene depths and fix colors concurrently from monocular underwater images to tackle the challenge of insufficient paired training data.

7.1.4 Segmentation

Segmentation divides an image into significant regions or objects, such as by separating seagrass from the adjacent water. This enables more accurate recognition of feature in monitoring applications. U-Net and DeepLab are frequently employed to segment seagrass patches accurately. Yang et al. (2020a, b) developed a two-dimensional membership matrix by utilizing the domain information of image pixels. They then determined the appropriate threshold by defining the criteria function as a two-dimensional multi-threshold fuzzy divergence. Liu et al. (2018) introduced a multi-threshold technique for image segmentation that integrated a search strategy with a differential evolution algorithm and gray Wolf algorithm. Morphological image segmentation is efficient and rapid. Ma et al. (2003) proposed a morphological filter group and segmentation technique based on texture features and contrast criteria to address the over-segmentation issue inherent in the watershed algorithm. Mariena et al. (2018) introduced a hybrid k-means clustering technique that incorporates the assessment of cluster centers. This technique addresses the significant impact of the initial number and position of cluster centers on segmentation efficacy. Cai and Jia (2009) introduced a fuzzy C-means clustering technique that incorporated spatial information by integrating a two-dimensional image histogram into

a conventional fuzzy C-means clustering algorithm and enhancing the membership function. Wang et al. (2010) introduced a technique that utilized an entropy constraint and rapid fuzzy C-means clustering, integrating the gradient operator, image histogram features, sampling calculations, and relative information loss for segmentation of underwater images. Sun et al. (2020) proposed an algorithm utilizing level-set methods that incorporate region and edge features for underwater image segmentation. The algorithm standardizes the level-set function through a distance regularization term, thereby improving the stability of its evolution and providing effective segmentation outcomes for underwater images. Li et al. (2010) redefined fuzzy entropy using the fuzzy entropy segmentation technique, employed an enhanced particle swarm optimization algorithm to determine the segmentation threshold, and proposed an underwater image segmentation system. Yan et al. (2020) used an asymmetric adaptive median filtering technique to enhance edge delineation between images' foreground and background, resulting in effective segmentation of underwater images.

7.2 Generation of Training Data

Data augmentation is commonly used to expand the training dataset for the classification of seagrass (Islam et al. 2018), particularly when the dataset is limited or lacks variety. Augmentation approaches are helpful when there is minimal diversity across seagrass images or where there is an imbalance among several classes. Standard augmentation techniques include modifying lighting conditions and applying rotations, flips, and transformations that simulate various environmental settings, thereby enhancing the generalization capacity of the model. Synthetic data, or artificially generated images, are essential when extensive annotated datasets are unavailable. Generating synthetic images in conjunction with actual data enhances the ability of the model to identify various properties in seagrass. These synthetic images can enhance the model's proficiency in classifying real-world seagrass correctly. Methods such as geometric adjustments, color alterations, and artificial noise simulate inherent variances in authentic underwater settings. Employing both augmented and synthetic data can enhance the reliability of seagrass classification models, thereby making them more suitable for real-time classification applications.

7.3 Data Labeling

Data labeling is essential in deep learning-driven classification of seagrass, as precisely annotated

samples improve the model performance and resilience. Labeling includes identifying different seagrass species, differentiating between seagrass and background elements, and annotating varying degrees of seagrass coverage and habitat classification. Accessibility and quality of labeled data are significant factors that influence the advancement and accuracy of deep learning models. Raine et al. (2020) created the 'DeepSeagrass' dataset, a patch-based multi-species seagrass collection in four- and five-class formats. They suggested a semi-automatic method for classifying the dataset, which reduced the manual labor. Noman et al. (2021a) automatically applied a semi-supervised deep learning framework employing EfficientNet-B5 to annotate extensive quantities of unlabeled multi-species seagrass data. This methodology enhances the classifier's accuracy by integrating labeled and pseudo-labeled data, producing exceptional performance on public and proprietary seagrass datasets. Moniruzzaman et al. (2019) developed a labeled dataset comprising 2699 underwater images of *Halophila ovalis*, with each image carefully annotated. Although hand labeling guarantees high precision, it is labor-intensive and requires considerable subject experience, especially when dealing with seagrass species exhibiting similar visual traits. The difficulty is exacerbated by fluctuating underwater lighting conditions, water turbidity, and insufficient contrast between the seagrass and background features. Gimenez-Romero et al. (2024) utilized a comprehensive georeferenced habitat dataset and a combination of satellite images to develop a robust model for mapping *Posidonia oceanica*, enhancing the efficiency of labeling and minimizing human labor. They used machine learning approaches to facilitate labeling, offering a scalable method for generating labeled data in extensive seagrass environments. Noman et al. (2024) presented an innovative Deep Neuroevolutionary (DNE) model that enhanced the labeling process. The model manages complex multi-species seagrass datasets adeptly based on a metaheuristic algorithm called 'Boosted Atomic Orbital Search (BAOS).' This efficient labeling method ensures precise model training and enhances generalization across diverse seagrass species. A summary of the datasets used in this study is presented in Table 2.

8 Detection Approaches

8.1 Seagrass Classification

Classification of seagrass is essential for its effective management and monitoring. Moniruzzaman et al. (2019) created a seagrass identification algorithm that employed a deep learning methodology. A labeled dataset comprising 2699 underwater photos of *Halophila ovalis* was generated using

a Faster R-CNN framework with an Inception V2 network for detection. This method attained a mean average accuracy (mAP) of 0.3464 on laboratory images and 0.261 on a mixed test set, highlighting the difficulties posed by the variability of underwater image, including minimal border differences and insufficient labeled datasets. Ozaeta et al. (2024) improved the identification of several seagrass species by assembling an in-depth view collection that reflected temporal fluctuations, diverse geographic locations, water depths, and turbidity levels. A deep learning methodology employing differentiable architecture search with early stopping was implemented to train a neural network, attaining an impressive 93.72% classification accuracy across five seagrass species: '*Cymodocea rotundata*,' '*Enhalus acoroides*,' '*Halophila ovalis*,' '*Syringodium isoetifolium*,' and '*Thalassia hemprichii*.' This highlights the capability of deep learning approaches to enhance the identification of seagrass species and facilitate more effective management of marine ecosystems.

8.2 Seagrass Mapping

Mapping of seagrass meadows is essential for efficient conservation and management. Conventional techniques for seagrass monitoring are frequently time-consuming, labor-intensive, and costly, constraining their scalability and applicability (Gimenez-Romero et al. 2024). The emergence of remote sensing and machine-learning technology has facilitated the development of more efficient and scalable monitoring systems. A comprehensive framework was developed to map *Posidonia oceanica* meadows in the Mediterranean Sea, addressing limitations such as limited availability of satellite imagery and insufficient model evaluation metrics by utilizing diverse satellite imagery and an extensive georeferenced habitat dataset to enhance model generalization across regions. Roelfsema et al. (2009) introduced an approach for mapping seagrass cover classes using field surveys and satellite data, particularly under the changing water conditions of Moreton Bay, Australia. They combined Landsat 5 Thematic Mapper images with in-situ data from video surveys and transects, effectively mapping seagrass at depths exceeding 3 m. This integration enabled the development of an extensive layer of polygons defining seagrass cover classes, providing dependable mapping for over 75% of the Bay, surpassing several prior efforts. Phinn et al. (2008) evaluated the accuracy of several remote sensing datasets, encompassing airborne hyperspectral and satellite multispectral images. The research indicated that precise mapping of seagrass characteristics across various image formats remains a problem. The dataset with the best mapping accuracy was obtained using aerial hyperspectral imaging (46%), followed by Quickbird-2 and Landsat-5.

Table 2 A summary of datasets used by authors in research papers

Dataset with reference	Type	Sensor and mounting vehicle	Data annotation	Data location	Class imbalance	Source
Proprietary seagrass dataset (Martin-Abadal et al. 2018)	Seagrass images	Not specified	Manual labeling	Indo-Pacific saltwater environments	Not specified	https://researchdata.edu.au/camris-seagrass-dataset/968386
LookingForSeagrass (Weidmann et al. 2019)	Underwater seagrass	Underwater camera	Not specified	Various global locations	Yes	https://github.com/EnviewFulda/LookingForSeagrass
Proprietary and DeepSeagrass (Noman et al. 2023)	Underwater and patch-based seagrass	Not specified	Combination of manual and automated labeling	Not specified	Yes	https://researchdata.edu.au/camris-seagrass-dataset/968386 and https://doi.org/10.25919/5f8e96f28f7d4
Underwater Halophila ovalis images (Moniruzzaman et al. 2019)	Underwater seagrass (Halophila ovalis)	Digital underwater camera	Manual annotation of Halophila ovalis leaves	Indo-Pacific region	Yes	https://ro.ecu.edu.au/datasets/41/
DeepSeagrass (Raine et al. 2020)	Patch-based multi-species seagrass	Not specified	Semi-automatic labeling for patches	Various marine locations	Yes	https://doi.org/10.25919/5f8e96f28f7d4
ECU-MSS (Noman et al. 2021a)	Patch-based multi-species seagrass	Not specified	Semi-supervised labeling	Not specified	Yes	https://ro.ecu.edu.au/datasets/141/
Posidonia oceanica dataset (Burguera 2020)	Seagrass cover mapping	Digital camera	Manual annotation of seagrass cover	Mediterranean Sea	Yes	https://datos.gob.es/en/catalogo/a04003003-posidonia-oceanica-provisional
Custom drone imagery (Jeon et al. 2021)	Coastal seagrass habitats	Drone-mounted optical	Manual annotation using a drone	Coastal areas	Yes	https://datasetninja.com/drone-dataset-uav
Global Wetlands (Raine et al. 2024)	Remote sensing data	Multi-spectral satellite and field surveys	Image-level labels	Wetlands across multiple regions	Yes	https://data.giss.nasa.gov/landuse/wetland.html
WorldView-2 images (Mederos-Barrera et al. 2022), (Manuputty et al. 2016) and (Baumstark et al. 2016)	High-resolution satellite imagery	Satellite-based imaging	Object-based classification, pixel-wise	Various locations (e.g., coastal, island regions)	No	https://earth.esa.int/eogateway/missions/worldview-2
Sentinel-2A satellite data (Bayyana et al. 2020)	Multi-spectral satellite imagery	Satellite-based imaging	Object-based annotation	Various coastal regions globally	Yes	https://www.satimagingcorp.com/satellite-sensors/other-satellite-sensors/sentinel-2a/
Landsat-8 (Topouzelis et al. 2018)	Multi-spectral satellite imagery	Satellite-based imaging	Object-based annotation	Global coastal regions	Yes	https://www.usgs.gov/landsat-missions/landsat-8
QuickBird satellite images (Barrell et al. 2015)	High-resolution satellite imagery	Satellite-based imaging	Object-based annotation, pixel-wise	Coastal and underwater regions globally	No	https://www.satimagingcorp.com/satellite-sensors/quickbird/

9 Learning Methods

9.1 Supervised Learning

Supervised learning involves training a model using a labeled dataset, wherein each input image is associated with a matching label. This method allows the model to learn mappings from inputs to outputs efficiently, addressing classification and regression issues effectively. Raine et al. (2020) created a multi-species seagrass detector and classifier using a deep convolutional neural network, achieving an overall accuracy of 92.4% in the classification of seagrass. The supervised learning architecture of the model distinguishes several seagrass species proficiently using labeled training data, improving the efficacy of classification and reducing the effort needed for data preparation.

9.2 Semi-supervised Learning

Semi-supervised learning combines supervised and unsupervised learning by utilizing labeled and unlabeled data. Noman et al. (2021b) introduced a semi-supervised two-stage approach that uses an extensive collection of unlabeled seagrass data and a limited labeled sample to enhance classification accuracy. The methodology achieved 98.0% accuracy in multi-species recognition on a newly developed seagrass dataset and, when utilized on the DeepSeagrass dataset, produced accuracies of 91.3% and 93.0%, respectively. Raine et al. (2024) applied a semi-supervised coarse segmentation method for seagrass classification minimize the labeling efforts. In contrast to conventional techniques that depend on patch-level supervision, their model employs image-level labels during training, thereby diminishing the necessity for extensive labeling by 25%. Two principal designs were introduced: SeaFeats, utilizing unsupervised contrastive pre-training for feature extraction, and SeaCLIP, which handles vast language models for domain-specific supervisory signals. The integration of SeaFeats

and SeaCLIP has shown enhanced efficacy, increasing the class-weighted F1 score on the 'DeepSeagrass' dataset by 6.8% and elevating the presence/absence F1 score on the 'Global Wetlands' dataset by 12.1%.

9.3 Unsupervised Learning

The lack of labeled training data defines unsupervised learning. Models attempt to identify patterns, structures, or distinct features in data without related annotations. This enables the model to categorize the input data into significant output classifications, effectively grouping similar data points. In the monitoring of seagrass, unsupervised learning is essential for the segmentation of habitats and the identification of unique seagrass features without manual labeling. Several studies have investigated the use of unsupervised learning for seagrass classification and habitat mapping. Jeon et al. (2021) investigated the efficacy of drone imaging and deep learning for the monitoring of seagrass. Their research evaluated several unsupervised learning models for the segmentation of seagrass in optical vision. Techniques such as U-Net, SegNet, PSPNet, and DeepLab v3+ were assessed, with U-Net exhibiting better performance despite its more straightforward design. The research highlighted the significance of normalizing techniques, indicating that min–max normalization was ideal for optical pictures, while Z-score normalization was most effective for black-and-white images. In another study, Abid et al. (2024) presented Unsupervised Curriculum Learning (UCL) for the seagrass classification. The UCL method achieved superior accuracy and recall rates, exceeding conventional supervised techniques such as SimCLR and unsupervised methods such as Zero-shot CLIP. The UCL method encompasses high-dimensional feature extraction, formation of pseudo-labels formation via clustering, accurate selection of samples, and iterative fine-tuning. Table 3 shows the comparative performance of various seagrass detection techniques using evaluation metrics (Table 4).

Table 3 Comparative performance of various seagrass detection methods using precision, recall, and F1-Score on grass and background segmentation tasks

Methods	Type	Labels	Grass			Background		
			Precision	Recall	F1- score	Precision	Recall	F1- score
Zero-short CLIP (Jeon et al. 2021)	Unsupervised	No labels	0.5974	0.4601	0.5192	0.7992	0.8723	0.8342
UCL	Unsupervised	No labels	0.9612	0.8606	0.9078	0.7989	0.9421	0.8633
SeaFeats + SeaCLIP (Raine et al. 2024)	Semi-supervised	Patch Image	0.9442	0.9662	0.9439	0.9579	0.9579	0.9425
EfficientNet-B5 (Noman et al. 2021b)	Supervised	Patch	0.8900	0.9367	0.9128	0.9600	0.8300	0.8903
VGG-16 with 2-Layer + Drop (Raine et al. 2020)	Supervised	Patch	0.9911	0.9866	0.9888	0.9699	0.9345	0.9083
SimCLR (Jeon et al. 2021)	Supervised	Patch	0.8188	0.9787	0.8916	0.9964	0.9801	0.9749

Table 4 The evaluation metrics applied by related works

No.	Performance metric	Meaning
1	Classification accuracy (CA)	The ratio of accurately predicted occurrences (including true positives and negatives) to the total number of examples
2	True positive (TP)	The number of occurrences accurately identified as belonging to the positive class
3	False positive rate (FPR)	The proportion of negative cases misclassified as positive (FP) relative to the total number of real negative instances
4	False negative rate (FNR)	The proportion of positive cases misclassified as negative (FN) relative to the total number of true positive instances
5	Specificity	The ratio of real negatives to all actual negatives, assessing the capacity to identify negative cases accurately
6	Precision	The quotient of genuine positives to the aggregate of forecasted positives ($TP / (TP + FP)$)
7	Recall (sensitivity)	The proportion of true positives relative to the overall count of real positives ($TP / (TP + FN)$)
8	F1 score	The harmonic mean of accuracy and recall offers a balanced assessment of both metrics
9	Mean Intersection over Union (mIoU)	The average overlap between predicted segmentation and ground truth across all classes
10	Confusion matrix	A matrix that presents the quantities of true positives, true negatives, false positives, and false negatives for assessing model efficacy
11	Intersection over Union (IoU)	The ratio of the overlap of predicted and actual areas to the union of these areas is commonly utilized in the identification and segmentation of objects
12	Mean square error (MSE)	The average of the squared deviations between predicted values and actual values reflects prediction accuracy
13	Root mean square error (RMSE)	The square root of the mean squared error offers a standard deviation of forecast inaccuracies

10 Deep Learning Architectures

Deep learning techniques have demonstrated robust performance in various automated image analysis tasks for detecting and classifying seagrass.

The selection of deep learning architecture significantly impacts model performance in the classification of seagrass. For example, compared to the baseline 88.2% accuracy shown by CNN, EfficientNet-B4 achieves higher accuracy of 93.7% on multi-species datasets due to its compound scaling mechanism, which optimizes resolution, depth, and width. In contrast, ResNet-50 demonstrates strong generalization across underwater video frames but underperforms in low-light conditions (F1-score drops to 0.82 in turbid water). Hybrid models like CNN-LTSM excel in the analysis of temporal data with 94.1% accuracy but require extensive training resources. These trade-offs highlight the need for architecture selection based on specific monitoring scenarios, such as prioritizing accuracy for conservation studies or computational efficiency for autonomous underwater vehicles (AUVs).

10.1 Convolutional Neural Network (CNN)

Martin-Abadal et al. (2018) developed a deep learning model utilizing VGG16-FCN8 for seagrass classification, with a precision of 96.57% and an accuracy of 96.81%. Weidmann et al. (2019) evaluated various deep learning models (DeepLabV3+, FCN, dilNet, U-Net) on the

'LookingForSeagrass' dataset, and determined that DeepLabV3+ attained the highest mean Intersection over Union (mIoU) of 87.78%. Wang et al. (2020) employed an encoder-decoder segmentation model on an identical dataset and achieved a remarkable mIoU of 90.66%. Moniruzzaman et al. (2019) developed a seagrass detection model utilizing a Faster R-CNN with an Inception V2 backbone, attaining a mean average precision (mAP) of 0.3464. Noman et al. (2021b) enhanced the efficacy of the model by employing a NASNet backbone and EfficientDet-D7 architecture within the Faster R-CNN framework, achieving mAP scores of 0.412 and 0.484, respectively. Jeon et al. (2021) compared various deep learning models (SegNet, U-Net, PSPNet, DeepLab v3+ to monitor seagrass ecosystems and found that U-Net performed better than the other models. Paul et al. (2023) presented a Lightweight Deep Learning System (LWDS) specially designed for seagrass classification. The LWDS obtained 93.52% accuracy for 250×250 images with $n = 4$, outperforming the standard approaches and cutting-edge techniques. The excellent accuracy, efficiency, and adaptability of LWDS make it a valuable tool for the classification of seagrass, surpassing other contemporary methods such as EfficientNetB5, Faster R-CNN, and conventional CNN models such as VGG-16, Xception, MobileNetV2, and SqueezeNet. Lestari et al. (2021) introduced a new mask region convolutional neural network (Mask R-CNN) algorithm to detect seagrass using underwater camera recordings. The proposed Mask R-CNN architecture could classify seagrass

correctly with a precision of 98.19% on the 'seagrass *Enhalus acoroides*' dataset. The results showed that Mask R-CNN detected and segmented the seagrass *Enhalus acoroides*' dataset with high efficiency. Tallam et al. (2023) conducted investigations using a cutting-edge image segmentation technique, DeepLab v3 model. High-resolution drone imagery collected over two years was used to train the model, demonstrating high accuracy in classifying eelgrass (temperate seagrass) across various conditions. Loureiro et al. (2024) analyzed several deep-learning algorithms for seabed characterization and habitat mapping. Their analysis emphasizes the benefits of deep-learning models, including CNNs, Generative Adversarial Networks (GANs), and Autoencoder (AE) Neural Networks, in managing complicated, high-dimensional data while enhancing classification accuracy and resilience. Deep-learning algorithms offer great potential for improving the quality and efficiency of seabed mapping, particularly in difficult underwater situations that present limited scope for data gathering. The author examined machine learning methods such as Support Vector Machine (SVM), Random Forest (RF), and k-nearest neighbor (k-NN), emphasizing their uses and limits. The study focused on using acoustic, magnetic, and optical sensors such as cameras. Although these techniques have been useful in various seafloor characterization tasks, they frequently fall short of scalability and show limited proficiency in dealing with the complexity of large-scale datasets compared to deep learning approaches. Mohamed et al. (2020b) proposed a semiautomated system for detecting and mapping benthic habitats and seagrass species based on convolutional neural networks (CNNs). The framework combines benthic habitat field data from a geo-located towed camera with high-resolution satellite images. Seagrass features derived from pre-trained CNNs were employed for detection, along with a "bagging of features" (BOF) technique. The model was evaluated using two study locations, Shiraho and Fukido (both situated on Ishigaki Island, Japan). The total accuracy in the detection of benthic habitats in Shiraho was 91.5%, whereas it was 90.4% for the detection of seagrass species in Fukido. Perez et al. (2020) introduced two deep learning models, the Convolutional Neural Network (CNN) and the Deep Capsule Network (DCN), for the autonomous qualifications of seagrass distribution with 8-band satellite images. The models were intended to optimize the classification and regression tasks collectively. This study demonstrated the efficiency of transfer learning in adapting models trained in one place to new locations with minimum extra input. Deep learning models are more accurate than typical machine learning approaches (linear regression and SVM). However, the study emphasized the need for global validation and enhanced ground truth data to improve the generalizability and accuracy of the model.

10.2 Transfer Learning

Raine et al. (2020) introduced the 'DeepSeagrass' dataset, employing VGG-16 and ResNet-50 classifiers through transfer learning to get accuracies of 88.2% for four classes and 92.42% for five classes. Noman et al. (2021a) improved the classification efficacy on the 'DeepSeagrass' dataset using the EfficientNet-B5 network, achieving 91.3% and 93% accuracies. They built a two-stage semi-supervised framework utilizing EfficientNet-B5 to autonomously annotate extensive datasets, achieving an accuracy of 98% on the proprietary 'ECU-MSS' dataset. Raine et al. (2024) proposed a new strategy for seagrass segmentation that relies solely on image-level labels and limited supervision. The technique combines the generic semantic understanding of numerous items of the CLIP model with domain-specific expertise to predict image seagrass patches. The two primary classifiers, SeaFeats and SeaCLIP, were merged into an ensemble to improve performance, particularly in visually damaged regions. The ensemble technique outperformed previous cutting-edge algorithms in addressing multi-species and binary seagrass classification challenges. The overall accuracy of SeaFeats + SeaCLIP was 96.04%.

10.3 Neural Network for Habitat Mapping

Burguera (2020) created two neural networks to map the *Posidonia oceanica* seagrass. The CNN-based model attained a precision of 97%, recall of 95.4%, and accuracy of 95.5%.

10.4 One-Stage and Two-Stage Detectors

Noman et al. (2023) demonstrated advanced methods for seagrass detection using YOLOv5-based one-stage and EfficientDetD7-based two-stage detectors, exhibiting strong detection efficacy using underwater images. One-stage and two-stage detection frameworks exhibit different trade-offs in accuracy and efficiency when applied to seagrass classification. One-stage detectors, such as YOLOv5, identify seagrass zones in efficient real-time, utilizing only one neural network pass. This reduces inference time and computational costs, making it appropriate for real-time applications. However, their classification accuracy may be significantly lower due to overlapping seagrass species and false positives in complicated underwater backgrounds, with a reported accuracy of around 94.5%.

In contrast, two-stage methods, such as the EfficientDetD7-based model, create region proposals and then refine classification, which makes them more accurate but slower. These models have a greater detection accuracy (~96.8%), since they minimize false positives efficiently. However, due to their two processes, they require more

computational power and take longer to process each image. The choice between one-stage and two-stage models is based on application requirements—YOLOv5 is better for real-time monitoring, while EfficientDetD7 is more suitable for high-precision classification.

10.5 Deep Neuroevolutionary (DNE) Approach

Noman et al. (2024) proposed a Deep Neuroevolutionary (DNE) model utilizing a novel metaheuristic method, 'Boosted Atomic Orbital Search (BAOS),' for the optimization of CNN architecture and hyperparameters. The BAOS-CNN model attained a remarkable overall accuracy of 97.48% on a multi-species seagrass dataset, with accuracies of 92.30% and 93.5% on the four-class and five-class variants of the 'DeepSeagrass' dataset, respectively.

11 Performance Evaluation Metrics for Seagrass Classification

In deep-learning-based classification of seagrass, assessing classifier performance is essential for measuring accuracy, generalization capabilities, and resilience. Various metrics fulfill distinct functions depending on the model type, classification task, and dataset. The primary evaluation standards were classification accuracy (CA), precision, recall, F1 score, mean Intersection over Union (mIoU), and mean average precision (mAP), as shown in Table 5.

11.1 Metrics Based on Accuracy and Precision

Numerous publications have focused on classification accuracy (CA) as a metric to assess the overall accuracy of a model's predictions. Martin-Abadal et al. (2018) evaluated their VGG16-FCN8 model for seagrass detection, attaining a precision of 96.57% and an accuracy of 96.81%. Burguera (2020) created a CNN-based model for *Posidonia oceanica*, attaining a precision of 97%, a recall of 95.4%, and an

accuracy of 95.5%. Raine et al. (2020) applied transfer learning utilizing VGG-16 and ResNet-50 classifiers, concentrating on the classification accuracy of four- and five-class iterations of the 'DeepSeagrass' dataset, attaining accuracies of 88.2% and 92.42%, respectively.

11.2 F1 Score, Recall, and Mean Intersection over Union Metrics

The F1 score, which balances accuracy and recall, is frequently used. Weidmann et al. (2019) used the mIoU metric to assess the efficacy of several deep learning models, including DeepLabV3+, FCN, and U-Net, based on the 'LookingForSeagrass' dataset. The DeepLabV3+ model attained an impressive mIoU score of 87.78%, underscoring its robust generalization ability. Wang et al. (2020) achieved a mIoU of 90.66% using an encoder-decoder segmentation model on an identical dataset. Noman et al. (2021b) achieved enhanced mAP using a Faster R-CNN framework with various topologies (NASNet and EfficientDet-D7), reaching mAPs of 0.412 and 0.484, respectively, demonstrating the model's efficacy in detecting seagrass in complex underwater settings.

11.3 Run-Time and Model Efficiency

In addition to classification measures, run-time performance and inference duration were assessed to evaluate the efficiency of various models. Jeon et al. (2021) conducted a comparative analysis of several deep learning models, highlighting the efficacy of U-Net, which attained superior accuracy while preserving its computational simplicity. The Deep Neuroevolutionary (DNE) model developed by Noman et al. (2024) showed an equilibrium between model complexity and performance, attaining an overall accuracy of 97.48% while automating architectural engineering and hyperparameter optimization.

Table 5 Deep learning architectures

Model	Dataset	Accuracy (%)	F1-score	Strengths	Limitations
CNN (baseline) (Noman et al. 2021a; b)	DeepSeagrass	88.2	0.85	Simple architecture, fast inference	Struggles with images captured in turbid water
EfficientNet B4 (Ozaeta et al. 2024)	Multi-species	93.7	0.92	Robust against noise, multi-scale feature extraction	High memory requirements
U-Net modified (Menghani 2023)	Satellite imagery	89.5	0.88	Pixel-wise segmentation	Requires dense annotations
ResNet-50 (Ozaeta et al. 2024)	Underwater video frames	91.3	0.89	Transfer-learning friendly	Sensitive to lighting variations
Hybrid CNN LSTM (He et al. 2023)	Temporal sensor data	94.1	0.93	Spatial-temporal dynamics	Computationally intensive

11.4 Use of Multiple Evaluation Metrics

The selection of evaluation criteria depends on the purpose of the study. Noman et al. (2021a) concentrated on semi-supervised learning and demonstrated the efficacy of their EfficientNet-B5 network in attaining excellent accuracy and resilience using the DeepSeagrass dataset. They reported an amazing accuracy of 98.0% on the ECU-MSS dataset using a two-stage framework, underscoring the significance of accuracy and model adaptability in integrating labeled and unlabeled data.

While metrics such as accuracy, precision, recall, and F1-score for performance evaluation provide a quantitative assessment of deep learning models, it is crucial to acknowledge their limitations in reflecting real-world applicability of the models for monitoring seagrass. One significant limitation is the dependence on the quality and representativeness of the datasets used for evaluation (Gangwal et al. 2024). In case the datasets do not adequately capture the environmental variability (e.g., different levels of water turbidity, lighting conditions, and seagrass densities) encountered in actual field deployments, the reported performance metrics may not accurately reflect the model's performance in operational scenarios (Adibnia et al. 2024). Furthermore, these metrics often treat all misclassifications equally, whereas, in practice, misclassifying a healthy seagrass meadow as a degraded one might have different ecological and management implications compared to the reverse. Therefore, future research should focus on incorporating more ecologically relevant evaluation metrics and conducting comprehensive validation studies in diverse field conditions to ensure the robustness and reliability of deep learning models for monitoring seagrass. The considerations of inference time and computational resources are also very important for real-time applications.

12 Challenges

Many challenges continue to exist in seagrass monitoring using deep learning techniques. A primary challenge is the dependence on high-quality labeled data, which is frequently limited in underwater settings because of the complicated process of obtaining accurate labels. Environmental variables such as water turbidity, lighting fluctuations, seasonal modifications, and image clarity affect the accuracy and performance of models. The significant similarity between seagrass species and other marine vegetation, such as algae and corals, results in possible misinterpretation. The physical variety and dense vegetation in seagrass meadows present challenges in classifying and mapping species adequately. Integrating multi-modal data, including satellites, UAVs, and underwater imaging, is a continuous

challenge, particularly with regard to generalizing models across varying environmental circumstances. Furthermore, although deep learning models have demonstrated potential, they are computationally demanding and require substantial processing power and data augmentation to reduce overfitting in smaller datasets. Achieving a balance between model complexity and performance while ensuring scalability is a significant challenge in operationalizing these technologies for large-scale monitoring. The efficacy of these models is often evaluated using standard metrics like accuracy and F1-score. However, these metrics can be misleading if the evaluation datasets do not represent the broad range of environmental conditions in which seagrass meadows exist. For example, a model that performs well on clear-water images might fail in turbid conditions. Therefore, it is crucial to interpret performance metrics in the context of the data used for evaluation and verify them through field validations.

Model architecture also introduces inherent limitations. For example, while CNNs are computationally efficient, they often fail to capture spatial-temporal relationships in video data, leading to misclassifications in dynamic underwater environments. Conversely, transformer-based architectures show promise for long-range dependency modeling but demand labeled datasets three times larger than CNNs, which are scarce in marine studies. Recent work by Ozaeta et al. (2024) demonstrates that EfficientNet variants outperform ResNet-50 by 6.2% in turbid conditions, underscoring the importance of environmental adaptability in model design.

13 Discussion

The use of deep learning models in seagrass monitoring has provided significantly enhanced outcomes compared with conventional techniques, offering the possibility for improved accuracy and efficiency in data processing. Models such as Convolutional Neural Networks (CNNs), Generative Adversarial Networks (GANs), and Deep Capsule Networks (DCNs) have exhibited exceptional efficacy in seagrass classification, primarily when utilized with datasets such as DeepSeagrass and proprietary collections. Transfer learning techniques have enhanced the adaptability of models, enabling models learned in one geographic area to be used in different locations with minimal supplementary data. Nonetheless, despite these advancements, problems with data availability, processing demands, and generalization across various marine settings persist. Recent developments, such as the Boosted Atomic Orbital Search (BAOS) for CNN optimization and semi-supervised learning frameworks, have enhanced the accuracy of classification and diminished dependence

on large labeled datasets. These solutions reduce the present limitations but need additional confirmation via real-world implementations. Moreover, unsupervised and semi-supervised learning methodologies have shown the potential to minimize labeling efforts and enhance performance in complex environments.

The survey's novelty lies in its hierarchical taxonomy, which categorizes deep learning models based on their architecture, data requirements, and performance, allowing for context-aware selection. Unlike previous surveys, this study integrated varied data insights to improve the accuracy and robustness of seagrass categorization. Our findings highlight the important need for models that balance accuracy, computing efficiency, and environmental flexibility, thereby directly addressing concerns of real-world applicability. It is expected the new identification of critical research gaps (e.g., domain adaptability, inadequate labeled data) will lead future investigations and facilitate AI-driven seagrass conservation.

14 Conclusion

This research survey effectively demonstrated the application of advanced deep-learning approaches for the accurate classification and mapping of seagrass habitats. We utilized information, including satellite and underwater imagery, and implemented models such as CNNs and transfer learning techniques to enhance the classification accuracy and efficiency in monitoring seagrass habitats. Our research addressed the issue of insufficiently labeled data by employing data augmentation approaches. This study examined the advantages of combining SVM with deep learning frameworks to attain an equilibrium between model complexity and performance. We attained elevated classification accuracy by comparing several algorithms, demonstrating that deep learning techniques can outperform conventional methods in complex maritime settings. The results of this study underscore the potential of automated, scalable seagrass monitoring systems that can assist in conservation and management efforts. These findings emphasize the necessity for future efforts to reinforce data generalization, real-time monitoring capabilities, and adaptation to diverse environmental situations. Furthermore, the novel taxonomy presented herein enables context-aware selection of models, paving the way for tailored deep learning applications that balance accuracy, efficiency, and environmental adaptability.

15 Future Research

Future research on seagrass monitoring should focus on improving data collection methodologies by automating the labeling process via unsupervised and semi-supervised learning approaches. This would reduce reliance on manually labeled data, which are frequently limited in aquatic contexts. Furthermore, a more comprehensive dataset can be obtained by combining multi-source data, including satellite, UAV, and underwater imagery, thereby enhancing model accuracy. Enhancements of cross-domain generalization, allowing models learned in one domain to be transferred to different contexts with minimal retraining, would be advantageous. Creating real-time monitoring systems with optimized neural networks can reduce the computational complexity and thereby improve the scalability and implementation of deep learning for seagrass monitoring. Hybrid models that integrate the advantages of conventional machine learning with deep learning methodologies can help reduce challenges such as class imbalance and data sparsity. Integrating explainable AI (XAI) methodologies enhances transparency and foster trust in model predictions. Algorithms that adapt to fluctuating environmental variables, including water turbidity and lighting fluctuations, should be developed to provide reliable performance in dynamic seagrass ecosystems. These developments are expected to enhance the efficacy and scalability of seagrass monitoring methods.

Funding Open Access funding provided by the Qatar National Library.

Data Availability DeepSeagrass, Global Wetlands, and ECU-MSS-2 datasets are publicly available and can be accessed through these links <https://arxiv.org/abs/2103.05226>, <https://www2.cifor.org/global-wetlands/>, <https://ro.ecu.edu.au/datasets/141/>.

Declarations

Conflict of interest The authors declare no conflict of interest.

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